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Roll No

EC-3004-CBGS

B.E. III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Electronic Devices and Circuits

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Discuss about Drift current and diffusion current.
b) Explain the working of PN junction diode in forward and reverse bias.
2. Draw the symbol and explain the working and application of following diodes.
 - i) Zener diode
 - ii) Varactor diode
 - iii) Schottky diode
 - iv) LED
3. a) Discuss the construction and basic operation of bipolar junction transistor.
b) Discuss about Eber's Moll model.
4. a) Discuss about voltage divider bias and collector to base bias of transistor.
b) Discuss the working of transistor as a switch.

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5. Discuss the analysis of transistor amplifier using h-parameters.
6. Discuss the working of following power amplifiers.
 - i) Class A
 - ii) Class AB
7.
 - a) Discuss the principle working of enhancement and depletion MOSFET.
 - b) Discuss the principle of operation of UJT.
8. Write short notes on any two of the following.
 - a) Tunnel diode
 - b) Hall effect
 - c) Silicon controlled rectifier

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